Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years building **customer-facing applications**, **REST APIs**, relational systems, and AI workflows using **Python**, **Java**, TypeScript, SQL, AWS, and rigorous testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Architected a multi-tenant AI education platform with **Azure SQL**, Blob Storage, row-level security, and normalized models for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Restructured fault-tolerant **AI workflows** using FastAPI background workers, processing 1K+ jobs with explicit state transitions and failure isolation while increasing throughput threefold under concurrent production workloads without cascading failures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Engineered a customer-facing grading platform using multimodal AI and OCR, evaluating 10K+ daily submissions with 90%+ consistency while delivering clear feedback workflows for more than 150 instructors.
- Designed concurrent **Python APIs** with secure file ingestion and AWS RDS persistence, processing 1K+ files per batch through fault-tolerant execution and CI/CD-backed releases with stable peak-load performance.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *LA, California*

- Built a Python Decision Transformer pipeline processing 100K+ trajectory records, improving long-horizon planning 18% through structured data preparation, repeatable model evaluation, simulation testing, and systematic experiment analysis.
- Architected a spatial-temporal encoder integrating HD-map geometry and historical agent trajectories through multi-head attention, improving closed-loop planning stability 15% while validating model behavior across diverse autonomous-driving scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed OOP modules with Singleton and Factory patterns for complex **API integrations**, improving response times 30%, increasing pipeline efficiency 68%, and reducing technical debt 20% through maintainable abstractions.
- Engineered cross-platform microservices with automated builds and 20+ **unit and integration tests** using Jenkins, PyTest, and Mockito, supporting reliable releases across three operating systems within Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Developed a **customer operations platform** using MERN, WebSockets, REST APIs, and eight microservices, synchronizing 2M+ records while reducing conflicts 33% across authentication, billing, permissions, and notification workflows.
- Automated Docker packaging and Jenkins CI/CD deployments, cutting release time 50%, while Azure SQL and analytics insights reduced resource waste 42% and increased customer lead conversion 15%.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++, C#
Backend & Data: REST APIs, FastAPI, Flask, Node.js, OOP, Data Structures, Data Modeling, PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB
Frontend: React, Next.js, HTML, CSS, TailwindCSS, Responsive Design, Customer-Facing Applications
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Monitoring
Quality & Practices: Unit Testing, Integration Testing, PyTest, Mockito, Mocking, Design Patterns, Code Reviews, Agile/Scrum, Privacy, Documentation

Projects

PointFlowLab | Python, SQLite, A/B Testing, Privacy

- Built and tested an **A/B offer workflow** with typed OOP services, deterministic assignment, relational persistence, conversion reporting, PII-minimized audit records, mocked dependencies, and positive and negative eligibility tests.

FinNexus Live | Next.js, TypeScript, Kubernetes, MongoDB

- Built a scalable **fintech platform** supporting 50K+ users and 1M+ transactions, integrating Plaid bank linking, token-based security, CI/CD, and real-time account updates aligned with regulated financial workflows.

Chat2SQL Code | Java, TypeScript, SQL, Docker

- Worked across a **Java** and TypeScript codebase with Docker, Maven, Node.js, and Yarn, supporting 10+ relational and NoSQL databases through schema exploration, query authoring, and cross-platform workflows.